Example Rubric

Question 1 (1 point)

Identify well-known algorithms that solve this optimization problem. Briefly describe each one and justify its suitability for meeting the company's requirements.

- Correctly identifies Prim's and Kruskal's algorithms as appropriate algorithms for this optimization problem. 0.4
- Avoids mentioning clearly inappropriate algorithms for this task, such as DFS or BFS as direct solutions to the optimization problem. 0.2
- Correctly explains Prim's algorithm as starting from a vertex and growing the minimum spanning tree by repeatedly selecting the minimum-weight adjacent edge. 0.15
- Correctly explains Kruskal's algorithm as constructing the minimum spanning tree by adding edges in increasing order of weight. 0.15
- Recognizes that cycles must not be formed in either approach. 0.05
- Correctly addresses the time complexity, avoiding incorrect claims such as stating that both algorithms are $O(n)$. 0.05

Question 2 (3 points)

Implement an algorithm that, given the optimized circuit obtained in the previous section, identifies the two pins whose distance, defined as the length of the shortest path between them, is maximum. The algorithm must return that distance and the intermediate pins forming the shortest path between the corresponding pair of pins. The code must include comments explaining how it works. Show an example of how to call the algorithm from a main program, specifying the initial parameter values.

- Correctly understands the objective of the problem: find the pair of pins whose shortest-path distance is maximum. 0.8
- Uses an appropriate shortest-path strategy to solve the problem. 0.7
- Correctly reconstructs and returns the intermediate pins forming the corresponding shortest path. 0.6
- Provides a coherent and functional implementation. 0.4
- Includes explanatory comments in the code. 0.2
- Includes an example of how to call the algorithm from a main program, specifying the initial parameter values. 0.3